

# **SYSTEM ADMINISTRATION AND MAINTAINANCE LAB REPORT**

Week 4 - File Systems and Storage Management

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## IT 123 – Week 4 Laboratory Report

File Systems and Storage Management — Lab Performance Documentation

|                                  |                                     |
|----------------------------------|-------------------------------------|
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### Ubuntu Server (Main Lab Disk)

#### A. Partition & Mount

- **Device Name:** /dev/sdb1
- **File System Type:** ext4
- **Mount Point:** /mnt/data

#### Detecting the New Disk

```
Ubuntu 26.04.1 LTS ubuntu-server tty1
ubuntu-server login: catlyn
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:   https://landscape.canonical.com
 * Support:      https://ubuntu.com/pro

System information as of Sat Sep  5 01:42:08 PM UTC 2026

System load:        0.65
Usage of /:         29.8% of 24.44GB
Memory usage:       9%
Swap usage:         0%
Processes:          102
Users logged in:    0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fea0:ca67

Expanded Security Maintenance for Applications is not enabled.

5 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

catlyn@ubuntu-server:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0  66.8M  1 loop /snap/core24/1643
loop1       7:1      0   1.5M  1 loop /snap/hwctl/123
loop2       7:2      0  50.1M  1 loop /snap/snapd/27710
sda         8:0      0   25G  0 disk
├─sda1       8:1      0    1M  0 part
├─sda2       8:2      0   25G  0 part /
sdb         8:16     0    5G  0 disk
sr0         11:0     1 1024M  1 rom
catlyn@ubuntu-server:~$
```

*Figure 1. New disk detected via lsblk.*



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Before starting the partitioning process, the `lsblk` command was used to check if the system could see the new virtual hard disk. The output showed a new 5GB disk named `/dev/sdb`, which was different from the existing `/dev/sda` system disk and did not yet have any partitions. This step was done to make sure the correct disk name was used in the next steps, since using the wrong disk could cause damage to the main system drive.

## Creating a New Partition

```
catlyn@ubuntu-server:~$ sudo fdisk /dev/sdb
[sudo: authenticate] Password:

Welcome to fdisk (util-linux 2.41.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x8d3b6426.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-10485759, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-10485759, default 10485759):

Created a new partition 1 of type 'Linux' and of size 5 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

catlyn@ubuntu-server:~$
```

*Figure 2. Partition created using fdisk.*

The command `sudo fdisk /dev/sdb` was used to create a new partition on the disk. The default options were selected so that the partition, named `/dev/sdb1`, would use the entire 5GB of space. After setting up the partition, the write command (`w`) was entered to save the changes. The result showed that the partition table was updated successfully, which meant the new partition was now ready to be formatted.



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## Formatting the Partition

```
catlyn@ubuntu-server:~$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 1310464 4k blocks and 327680 inodes
Filesystem UUID: cc096718-c488-457f-992a-b36029ef4720
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

catlyn@ubuntu-server:~$
```

*Figure 3. Partition formatted with ext4.*

After creating the partition, the command `sudo mkfs.ext4 /dev/sdb1` was used to format it using the ext4 file system, which is commonly used on Linux systems. The output showed the number of blocks and inodes created, and confirmed that the file system was successfully set up. This step prepared the partition so it could be used to store files and folders.

## Mounting the Partition

```
catlyn@ubuntu-server:~$ sudo mkdir /mnt/data
catlyn@ubuntu-server:~$ sudo mount /dev/sdb1 /mnt/data
catlyn@ubuntu-server:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           680M  1.1M  679M   1% /run
/dev/sda2       25G   7.4G   16G  32% /
tmpfs           1.7G   0  1.7G   0% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.7G   0  1.7G   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           340M   8.0K  340M   1% /run/user/1000
/dev/sdb1       4.9G  1.3M  4.6G   1% /mnt/data
catlyn@ubuntu-server:~$
```

*Figure 4. Partition mounted at /mnt/data.*

A folder named `/mnt/data` was created using `sudo mkdir` to act as the mount point. The formatted partition `/dev/sdb1` was then mounted to this folder using `sudo mount`. Running `df -h` afterward showed that `/dev/sdb1` was now listed and mounted at `/mnt/data`, with about 4.6GB of space available. This confirmed that the disk was ready to be used and could now be configured with permissions.



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## B. Directory Setup & Permissions

- **Directory Path:** /mnt/data
- **Group Assigned:** facultygrp (owner group) — studentgrp (no access)
- **Permission Level:** 770 (rwxrwx---) — root:facultygrp full access, all others none

## Creating Groups and User Accounts

```
catlyn@ubuntu-server:~$ sudo groupadd facultygrp
catlyn@ubuntu-server:~$ sudo groupadd studentgrp
catlyn@ubuntu-server:~$ sudo useradd -m -G facultygrp faculty2
catlyn@ubuntu-server:~$ sudo useradd -m -G studentgrp student4
catlyn@ubuntu-server:~$ sudo passwd faculty2
New password:
Retype new password:
passwd: password updated successfully
catlyn@ubuntu-server:~$ sudo passwd student4
New password:
Retype new password:
passwd: password updated successfully
catlyn@ubuntu-server:~$ _
```

*Figure 5. Groups and test user accounts created.*

Two groups, facultygrp and studentgrp, were created using sudo groupadd to represent the two types of users who would need different access levels. Two test accounts, faculty2 and student4, were then created using sudo useradd, which also made their home folders and placed each user in their proper group. Finally, sudo passwd was used to set a password for each account, which was needed to log in and test their access later using the su command.

## Setting Ownership and Permissions

```
none          1.0M      0  1.0M    0% /run/credentials/systemd-networkd.service
none          1.0M      0  1.0M    0% /run/credentials/getty@tty1.service
tmpfs         340M     8.0K   340M    1% /run/user/1000
/dev/sdb1     4.9G     1.3M   4.6G    1% /mnt/data
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$ sudo chown root:facultygrp /mnt/data
catlyn@ubuntu-server:~$ sudo chmod 770 /mnt/data
catlyn@ubuntu-server:~$
```

*Figure 6. Ownership and permissions applied.*





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The command `sudo chown root:facultygrp /mnt/data` was used to change the group owner of the folder to `facultygrp`, while `root` remained the file owner. Right after, `sudo chmod 770` was applied, which gives full read, write, and execute access to both `root` and `facultygrp`, while removing all access for everyone else, including `studentgrp`. This setup made sure only members of `facultygrp` could use the folder.

## Verifying Permissions

```
catlyn@ubuntu-server:~$  
catlyn@ubuntu-server:~$  
catlyn@ubuntu-server:~$  
catlyn@ubuntu-server:~$ ls -l /mnt/  
total 4  
drwxrwx--- 3 root facultygrp 4096 Sep  5 14:11 data  
catlyn@ubuntu-server:~$
```

*Figure 7. Permissions verified with ls -l.*

To check if the previous `chown` and `chmod` commands worked correctly, `ls -l /mnt/` was run to show the details of the `/mnt` folder. The result showed `drwxrwx---` as the permission setting for the `data` folder, along with `root` and `facultygrp` listed as the owner and group. This confirmed that the permission setup was correct before testing access using the faculty and student accounts.

## C. Verification

- **Tested User Account:** `faculty2` (write test) and `student4` (write test)
- **Access Allowed:** `faculty2` — ☒ Yes ☐ No → (successfully created file)
- **Access Denied:** `student4` — ☒ Yes ☐ No → ("Permission denied")

## Testing Faculty Account Access

```
catlyn@ubuntu-server:~$ su - faculty2  
Password:  
$ cd /mnt/data  
$ touch faculty_test.txt  
$ ls -l  
total 16  
-rw-rw-r-- 1 faculty2 faculty2 0 Sep  5 14:17 faculty_test.txt  
drwx----- 2 root      root    16384 Sep  5 14:11 lost+found  
$ exit
```

*Figure 8. Faculty account access test.*



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To test if the permission setup worked, the `su -` command was used to log in as `faculty2`, and a test file was created inside `/mnt/data` using `touch`. This worked without any errors, showing that `faculty2` has write access to the folder.

## Testing Student Account Access

```
catlyn@ubuntu-server:~$ su - student4
Password:
$ touch /mnt/data/student_test2.txt
touch: setting times of '/mnt/data/student_test2.txt': Permission denied
$ exit
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
```

*Figure 9. Student account access test.*

The same test was then done as `student4`, but this time the `touch` command gave a "Permission denied" message, showing that `student4`, who is not part of `facultygrp`, was correctly blocked from writing to the folder. These results proved that the permission setup worked as expected.

## Part 2 – Student Exercise (10GB Disk, /projectdata)

### A. Partition & Mount

- **Device Name:** `/dev/sdc1`
- **File System Type:** `ext4`
- **Mount Point:** `/projectdata`

### Detecting the New 10GB Disk

```
catlyn@ubuntu-server:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0   1.5M  1 loop /snap/hwctl/123
loop1       7:1      0  66.8M  1 loop /snap/core24/1643
loop2       7:2      0  50.1M  1 loop /snap/snapd/27710
sda         8:0      0   25G   0 disk
├─sda1      8:1      0    1M   0 part
├─sda2      8:2      0   25G   0 part /
└─sdb       8:16     0    5G   0 disk
   └─sdb1    8:17     0    5G   0 part
sdc         8:32     0   10G   0 disk
sr0        11:0     1 1024M   1 rom
catlyn@ubuntu-server:~$
```

*Figure 9. New 10GB disk detected via `lsblk`.*



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After shutting down the VM and adding a new 10GB virtual hard disk through VirtualBox settings, the system was turned back on and lsblk was run to check if the disk was detected. The output showed the new disk as /dev/sdc, which was different from the /dev/sda system disk and the /dev/sdb disk used earlier for /mnt/data, and it did not have any partitions yet. This confirmed that the correct, unused disk was ready to be set up for the student exercise.

## Creating a New Partition

```
catlyn@ubuntu-server:~$ sudo fdisk /dev/sdc
[sudo: authenticate] Password:

Welcome to fdisk (util-linux 2.41.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xc640cdbc.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-20971519, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971519, default 20971519):

Created a new partition 1 of type 'Linux' and of size 10 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

catlyn@ubuntu-server:~$
```

*Figure 10. Partition created on the 10GB disk.*

The command `sudo fdisk /dev/sdc` was used to create a new partition on the disk. The default settings were used so that the partition, named /dev/sdc1, would take up the entire 10GB of space. After choosing the partition type and accepting the default settings, the write command (w) was used to save the changes. The output showed that the partition table was updated, meaning the disk was now ready to be formatted.





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## Formatting the Partition

```
catlyn@ubuntu-server:~$ sudo mkfs.ext4 /dev/sdc1
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 2621184 4k blocks and 655360 inodes
Filesystem UUID: d884117c-0202-44ef-be11-6d7bf5efaea0
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

catlyn@ubuntu-server:~$
```

*Figure 11. Partition formatted with ext4.*

The new partition `/dev/sdc1` was formatted using `sudo mkfs.ext4`, which set up the ext4 file system needed to store data. The output showed the number of blocks and inodes created for the 10GB partition, and confirmed that the file system was set up successfully. This step prepared the disk to be mounted and used as the `/projectdata` folder.

## Mounting the Partition

```
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$ sudo mkdir /projectdata
catlyn@ubuntu-server:~$ sudo mount /dev/sdc1 /projectdata
catlyn@ubuntu-server:~$ df -h /projectdata
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdc1        9.8G  2.1M  9.3G   1% /projectdata
catlyn@ubuntu-server:~$
```

*Figure 12. Partition mounted at /projectdata.*

A new folder, `/projectdata`, was created using `sudo mkdir`, and the formatted partition `/dev/sdc1` was mounted to it using `sudo mount`. Running `df -h /projectdata` afterward showed that the partition was mounted successfully, with about 9.3GB of the 9.8GB total space available. This confirmed that the disk was properly connected and ready for permission setup.



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## B. Directory Setup & Permissions

- **Directory Path:** /projectdata
- **Group Assigned:** facultygrp (read/write) — studentgrp (read-only via ACL)
- **Permission Level:** 770 base (root:facultygrp rwx) + ACL: studentgrp r-x

## Applying Permissions and ACL

```
catlyn@ubuntu-server:~$ sudo apt install acl -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30      linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Installing:
  acl

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 5
  Download size: 39.4 kB
  Space needed: 197 kB / 17.0 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu/resolute/main amd64 acl amd64 2.3.2-2 [39.4 kB]
Fetched 39.4 kB in 0s (115 kB/s)
Selecting previously unselected package acl.
(Reading database... 139547 files and directories currently installed.)
Preparing to unpack .../archives/acl_2.3.2-2_amd64.deb...
Unpacking acl (2.3.2-2)...
Setting up acl (2.3.2-2)...
Processing triggers for man-db (2.13.1-1build1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

*Figure 13. Mixed permissions applied using base permissions and ACL.*

Since the task required faculty to have read and write access while students only had read access, using chmod alone was not enough to meet both needs at the same time. The acl package was first installed using `sudo apt install acl -y` to allow more detailed permission settings. Ownership was then set to root:facultygrp with `sudo chown`, and `sudo chmod 770` was applied as the base permission. Finally, `sudo setfacl -m g:studentgrp:r-x /projectdata` was used to add an extra rule that gives studentgrp read and execute access without letting them write, adding read-only access on top of the base permissions.



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## Verifying ACL Permissions

```
catlyn@ubuntu-server:~$ ls -ld /projectdata
drwxrwx---+ 3 root facultygrp 4096 Sep  5 14:26 /projectdata
catlyn@ubuntu-server:~$ getfacl /projectdata
getfacl: Removing leading '/' from absolute path names
# file: projectdata
# owner: root
# group: facultygrp
user::rwx
group::rwx
group:studentgrp:r-x
mask::rwx
other::---
```

*Figure 14. ACL permissions verified with getfacl.*

To check if the ACL setting was applied correctly, `ls -ld /projectdata` was run first, and a plus sign (+) appeared at the end of the permission text, showing that extra ACL rules were added on top of the normal permissions. The `getfacl /projectdata` command was then used to show the full list of access rules, which confirmed that `facultygrp` has full (rwx) access while `studentgrp` only has read and execute (r-x) access, without write. This confirmed that the mixed-permission setup worked as intended.

## C. Verification

- **Tested User Account:** `faculty2` (write test) and `student4` (read + write test)
- **Access Allowed:** `faculty2` — [✓] Yes [ ] No → (write) | `student4` — (read/list only)
- **Access Denied:** `student4` — [✓] Yes [ ] No → (write denied: "Permission denied")

## Testing Faculty and Student Access

```
catlyn@ubuntu-server:~$ su - faculty2
Password:
$ touch /projectdata/faculty_test.txt
$ ls -l /projectdata
total 16
-rw-rw-r-- 1 faculty2 faculty2  0 Sep  5 14:35 faculty_test.txt
drwx----- 2 root      root    16384 Sep  5 14:26 lost+found
$ exit
catlyn@ubuntu-server:~$ su - student4
Password:
$ ls -l /projectdata
total 16
-rw-rw-r-- 1 faculty2 faculty2  0 Sep  5 14:35 faculty_test.txt
drwx----- 2 root      root    16384 Sep  5 14:26 lost+found
$ touch /projectdata/student_test.txt
touch: cannot touch '/projectdata/student_test.txt': Permission denied
$ exit
catlyn@ubuntu-server:~$
```

*Figure 15. Read/write access confirmed for faculty and read-only for students.*



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Using `su - faculty2`, a test file was created inside `/projectdata` with `touch`, and the command worked without any errors, showing that `faculty2` has full write access to the folder. Switching to `student4` using `su -`, the `ls -l /projectdata` command was run first and successfully showed the contents of the folder, proving that students do have read access as planned with the ACL. However, when `student4` tried to create a file using `touch`, the system showed "Permission denied," confirming that students only have read-only access and cannot add or change files in the shared project folder.

## Part 3 – Reflection

### 1. What challenges did you encounter in creating partitions and managing permissions?

- One of the challenges we encountered was making sure we were working on the correct disk and partition, since commands like `fdisk` and `mkfs.ext4` can cause data loss if used on the wrong device. We had to be extra careful in checking the output of `lsblk` before proceeding with each step. We also ran into a few typing errors while creating users and groups, such as accidentally typing "eseradd" instead of "useradd," which taught us how to properly check command outputs and, when needed, delete and recreate users or groups to keep our setup clean. Another challenge was understanding how Linux permissions work, especially when the activity required mixed access levels, where `faculty` needed full access but students needed read-only access to the same folder. A basic `chmod` command was not enough to satisfy both requirements at once, so we had to learn how to use `setfacl` to add an extra permission rule on top of the basic owner-group-others system, which was a new concept for us.

### 2. How can these skills be applied in real-world system administration duties?

- These skills are directly applicable to real-world system administration, since managing storage devices and controlling who can access certain files and folders is one of the most common responsibilities of a system administrator. In an actual workplace, a sysadmin might need to set up shared network storage for different departments, making sure that each group only has access to what they are authorized to see or edit, similar to how we configured `facultygrp` and `studentgrp` in this activity. The use of tools like `fdisk`, `mkfs`, and `mount` also reflects how real servers are set up when new physical or virtual drives are added to expand storage capacity. Additionally, learning to use ACLs to apply more detailed and flexible permissions is especially useful in real organizations, where access requirements are often more complex than a single owner-group-others structure can handle.